



Witness data magic

In this video, I'm going to share
some case studies that
highlight the incredible work data analysts do.
Each of these scenarios shows off the power of
data-driven decision-making in unexpected ways.
The first story is about Google.
As I mentioned a little while back, here at Google,
our mission is to organize
the world's information and make it
universally accessible and useful.
All of our products,
from idea to development to launch,
are built on data and data-driven decision-making.
There are tons of examples here at Google of
people using facts to create business strategy.
But one of the most famous ones
has to do with Google's human resources.
Here's how it went. The HR department
wanted to know if there was value in having managers.
Were their contributions worthwhile?

Or should everyone just be an individual contributor?

To answer that question,

Google's people analytics team looked
at past performance reviews and employee surveys.

The data they found was plotted on
a graph because as you've learned,
visuals are extremely helpful
when trying to understand a problem or concept.

The graph revealed that Googlers
had positive feelings about their managers,
but the data was pretty general
and the team wanted to learn more.

So they dug deeper and split the data into **quartiles**.

A quartile divides data points into

four equal parts or quarters.

Here's where the really cool stuff started happening.

The data analysts discovered
that there was a big difference
between the very top and the very bottom quartiles.

As it turned out, the teams with
the best managers were significantly happier,
more productive, and more
likely to want to keep working at Google.

This confirmed that managers were
valued and make a big difference.

Therefore, the idea of having

only individual contributors was not implemented.

But there was still more work to do.

Just knowing that great managers create great results doesn't lead to actionable insights.

You have to identify what exactly makes a great manager, so the team took

two additional steps to collect more data.

First, they launched an awards program where employees could nominate their favorite managers.

For every submission you had to provide examples or data about what makes that manager great.

The second step involved interviewing managers who were graphed on the top and bottom quartiles.

This helped the analytics team see the differences between successful and less successful management behaviors.

The best behaviors were identified as were the most common reasons for a manager needing improvement.

The final step was sharing these insights and putting a procedure in place for evaluating managers with these qualities in mind.

This data-driven decision continues to create an exceptional company culture for

my colleagues and me. Thanks, data.

Another interesting example comes
from the nonprofit sector.

Nonprofits are organizations dedicated to advancing
a social cause or advocating for a particular effort,

such as food security,
education or the arts.

In this case, data analysts
researched how journalists can make
a more meaningful impact
for the nonprofits they would write about.

Because journalists write for newspapers,
magazines, and other news outlets,
they can help nonprofits reach readers like you and me,
who then take action to
help nonprofits reach their goals.

For instance, say you read about the problem
of climate change in an online magazine.

If the article is effective,
you'll learn more about the cause and might even be
compelled to make greener choices
in your day-to-day life,
volunteer for a nonprofit,
or make a donation.

That's an example of
the journalist's work bringing about awareness,

understanding, and engagement. So, back to the story.

The data analysts used a tracker to monitor story topics, clicks, web traffic, comments, shares and more.

Then they evaluated the information to make recommendations for how the journalists could do their jobs even better.

In the end, they came up with some great ideas for how nonprofits and journalists can motivate people everywhere to work together and make the world a better place.

There's really no end to what you can do as a data analyst.

As you progress through this program, you'll discover even more possibilities.

Great job following along with the topics of these past few videos.

You learned all about analytical skills and the five key characteristics of data analysts.

You probably even learned that you are a pro at most of these already.

Next, you discovered what it means to think analytically and the specific skills data analysts develop to help them do it.

You explored tools and processes that enable data analysts to

pinpoint a problem and ask

the right questions in order to solve them.

Finally, some real-world stories helped illustrate

why data-driven decision-making is

usually more successful than other methods.

You're building a wonderful foundation

for your career as a data analyst.

With every video, your skills will continue to expand, and

your understanding of key data analytics

concepts will only get stronger.

Soon, you'll have a chance

to test out everything you've learned.

This is a really useful opportunity to check

your understanding of all the concepts we've discussed,

and if you're ever unsure about a question,

you can review the videos and

readings to find the answer.

This is another awesome way to practice collecting data.

Keep up the great work.